

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	01 October 2026
Team ID	SWTID-2026-2085
Project Name	Import Data Using Transform Maps (Spreadsheet)
Maximum Mark	4 Marks

Functional Requirements :

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration and Login	Registration through form (name, email, password) Registration through Gmail Login with role-based access (Admin / User)
FR-2	User Confirmation	Confirmation via email Confirmation via OTP
FR-3	Spreadsheet Upload	Upload .xlsx / .csv files Validate file type and size Load the file into a staging (import set) table Preview the uploaded data before import
FR-4	Transform Map Creation	Select the source (staging) table and the target table Auto-map matching columns Manually map source fields to target fields Set a coalesce field to prevent duplicates
FR-5	Data Transformation and Import	Run the transform on staged data Validate data (required fields, data types, formats)
		Insert new records and update existing ones
FR-6	Import Results and Error Handling	View import summary (inserted, updated, failed counts) View error log for failed rows

		View import history
--	--	---------------------

Non-functional Requirements

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The interface is simple and easy to follow, with a clear upload, map, and import flow. Data preview and auto-mapping reduce effort for non-technical users.
NFR-2	Security	Passwords are hashed and access uses token-based authentication (JWT). Role-based access limits imports and map creation to authorized users, and uploaded files are validated to block malicious content.
NFR-3	Reliability	Failed rows are logged without stopping the entire import. Staging keeps the original data intact, and coalesce rules prevent duplicate or corrupted records.
NFR-4	Performance	Files with thousands of rows are processed within acceptable time, with target import time under [X] seconds for [N] records. Batch processing keeps memory use low.
NFR-5	Availability	The application is available around the clock (target [99%] uptime) when hosted on a cloud platform, with database backups for recovery.
NFR-6	Scalability	The system can handle growing file sizes and more concurrent users. Background job queues and database indexing support this, and new data sources or target tables can be added without redesign.